Tan Dai Ngo

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EDUCATION

University of Victoria | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Coursework:** Optimization for Machine Learning, Systems for Massive Datasets, Data Mining. **GPA:** 8.78 / 9.00.

University of Chicago | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Coursework:** Algorithms, Distributed Systems, Databases, Discrete Mathematics, Applied Software Engineering.

University of Washington | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics, Dean's List. **GPA:** 3.57 / 4.00

EXPERIENCE

Software Developer | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Developed and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and deployed features via Jenkins across 50+ releases.
- Tuned Cassandra queries to respond in under 1 second and debugged production issues with Splunk.

Math and Computer Science Teacher | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Explained technical material to 50+ students with no prior background, adjusting the pitch in real time.
- Mentored students one-on-one on debugging and algorithmic reasoning by working through their own broken code.

PROJECTS

Multi-Stage IoT Intrusion Detection | Python, XGBoost, PyTorch *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six supervised classifiers on 206,000 CIC IoT-DIAD 2024 flows with preprocessing fit on training only.
- Selected binary XGBoost under a 1% false positive budget, reaching 99.5% recall at 0.94% FPR and 0.989 PR-AUC.
- Designed a four condition holdout that exposed capture session leakage, dropping PR-AUC from 0.981 to 0.125.

Ocean Data ML Platform with RAG | Python, Spark, XGBoost, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Harmonized 10.8 million NOAA and Ocean Networks Canada records into Bronze, Silver and Gold layers with lineage.
- Diagnosed the data as the limit on water temperature while the same pipeline cut air temperature RMSE by 48%.
- Served semantic search behind FastAPI, Qdrant and Ollama at hit@k 0.9 and term recall 0.85 on held out queries.

End-to-End Multi-Output Fuel Blending System | Python, CatBoost, FastAPI, AWS *Shell.ai Hackathon 2025 | Jul 2025*

- Built and deployed multi output regression across 10 blend properties through FastAPI on Dockerized AWS EC2.
- Advanced CatBoost to deployment based on 5-fold cross-validation with a MAPE of 0.64 versus XGBoost's 1.29.
- Engineered entropy based mixture metrics and serialized preprocessing, so training and inference stay consistent.

Automated Crypto Trading Bot | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine that buys 1% dips and sells on 2% rises inside rolling 60 minute windows.
- Added a trend strategy that buys only when the 8 period EMA is above the 20 and price is above the 200 EMA.
- Backtested both on four months of polled data, then deployed to AWS EC2 with per market caps from a CSV config.

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, Go, Cypher, MATLAB, JavaScript, HTML, CSS
Machine Learning & Evaluation: PyTorch, XGBoost, CatBoost, scikit-learn, NumPy, Optuna, MLflow, Sentence Transformers
Data, Retrieval & Messaging: Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, MongoDB, Kafka
Backend, Cloud, CI/CD & Testing: Spring Boot, FastAPI, REST, OpenAPI, Docker, AWS (EC2, S3, Lambda), Git, Jenkins, pytest
Credentials: Google Advanced Data Analytics (2025), Shell.ai Hackathon 2025, IMC 64 Bids (top 250 of 1,500)